

DBMS – Week 3

Seller and Inventory Management System

1. Introduction

The **Seller and Inventory Management System** is used in an e-commerce company to manage sellers, products, and inventory.

The system maintains seller details, tracks products supplied by sellers, and monitors stock levels to ensure product availability.

2. Create Seller and Inventory Tables

The first step is to create two tables:

1. Seller Table

2. Inventory Table

These tables store seller information and product stock information.

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25),Email VARCHAR(30), Phone number(10), Address VARCHAR(45));
Table created.
SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR(7),
2 Product_ID VARCHAR(7), Seller_ID VARCHAR(7),
3 Stock_Quantity number, Stock_status VARCHAR(11),
4 Last_Updated DATE);
Table created.
```

Primary Key added for Both Tables:

Primary Key

Definition

A **Primary Key** uniquely identifies each record in a table.

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);  
Table altered.  
  
SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);  
Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_ID);  
Table altered.
```

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);  
Table altered.
```

Constraints

Constraints are rules applied to table columns to maintain correct data.

A. Unique Email

To prevent duplicate email addresses:

ALTER TABLE Seller

ADD CONSTRAINT UQ_SELLER_EMAIL UNIQUE (Email);

```
SQL> ALTER TABLE Seller ADD  
2 CONSTRAINT Un_mail UNIQUE(Email);
```

B. Email NOT NULL

MODIFY Email VARCHAR2(30) NOT NULL;

Email must contain a value:

ALTER TABLE Seller

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;  
Table altered.
```

C. Positive Stock Quantity

Stock quantity should not be negative:

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);  
  
Table altered.
```

Seller Table

Definition

The **Seller table** stores information about sellers who supply products to the e-commerce company.

Columns

Column	Description
Seller_ID	Unique ID of seller
Seller_Name	Name of seller
Email	Seller's email
Phone	Seller's phone number
Address	Seller's address

Create Seller Table

```
SQL> DESC Seller;
```

Name	Null?	Type
SELLER_ID	NOT NULL	VARCHAR2(10)
SELLER_NAME		VARCHAR2(20)
EMAIL	NOT NULL	VARCHAR2(30)
PHONE		NUMBER(38)
ADDRESS		VARCHAR2(50)

```
SQL> SELECT * FROM Seller;
```

SELLER_ID	SELLER_NAME	EMAIL	PHONE
S001	Arun Crafts	arun@gmail.com	9876543210
S002	Meena Arts	meena@gmail.com	9876543211
S003	Kavi Creations	kavi@gmail.com	9876543212
S004	Riya Designs	riya@gmail.com	9876543213

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',
2 &Stock,'&Stock_status',TO_DATE('&Last_upd','YYYY-MM-DD'));

Enter value for invid: Inv0001
Enter value for prodid: P100001
Enter value for sell: S00001
old 1: INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',
new 1: INSERT INTO Inventory VALUES('Inv0001','P100001','S00001',
Enter value for stock: 4
Enter value for stock_status: Available
Enter value for last_upd: 2026-09-12
old 2: &Stock,'&Stock_status',TO_DATE('&Last_upd','YYYY-MM-DD'))
new 2: 4,'Available',TO_DATE('2026-09-12','YYYY-MM-DD'))

1 row created.
```

```
SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
-----
Inv0001 P100001 S00001          4 Available  12-SEP-26
Inv0002 P100001 S00002          3 Available  13-AUG-26
Inv0003 P100002 S00003          4 Unavailable 15-SEP-26
Inv0004 P200002 S00004          3 Available  14-SEP-26
```

4. Seller and Product Relationship

A seller can supply multiple products.

A product can be supplied by one or multiple sellers.

Relationship

Seller → Product

1 Many

This means:

One Seller → Many Products

5. Product and Inventory Relationship

Each product has inventory information.

Inventory is used to track the available stock of products.

Relationship

Product → Inventory

1 1

This means:

One Product → One Inventory record

Report 1 – Seller-wise Product Report

Purpose

This report shows:

- Seller name
- Number of products supplied
- Available stock

Query

Because your Inventory table already contains SELLER_ID, you can join **Seller and Inventory**:

Report 2 – Stock Availability Report

Purpose

This report displays:

- Product name
- Stock quantity
- Stock status

Your current database does **not have a Product table**, so the PDF's Product_Name cannot currently be displayed.

You can still display the Product ID, stock quantity, and status:

Report 3 – Inventory Analysis

The report finds:

1. Total products available
2. Products out of stock
3. Highest stocked products
4. Average inventory quantity

A. Total Products Available

```
SQL> SELECT COUNT(*) AS Total_Available_Products
  2  FROM Inventory
  3  WHERE Stock_Status = 'Available';

TOTAL_AVAILABLE_PRODUCTS
-----
                        2
```

B. Products Out of Stock

Your table uses Unavailable as the stock status:

```
SQL> SELECT COUNT(*) AS Total_Available_Products
  2  FROM Inventory
  3  WHERE Stock_Status = 'Available';

TOTAL_AVAILABLE_PRODUCTS
-----
                        2
```

C. Highest Stocked Products

Use your actual column name STOCK_QUANTITY:

```
SQL> SELECT p.Product_Name, i.Stock_Quantity
  2  FROM Product p
  3  JOIN Inventory i ON p.Product_ID = i.Product_ID
  4  WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);

PRODUCT_NAME                                STOCK_QUANTITY
-----
Laptops                                     4
Head Phones                                4
```

D. Average Inventory Quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory  
2 FROM Inventory;
```

```
AVERAGE_INVENTORY  
-----  
3.5
```

Conclusion

The **Seller and Inventory Management System** helps an e-commerce company maintain seller information, track products supplied by sellers, and monitor inventory levels. Primary keys, foreign keys, and constraints maintain data integrity, while reports provide information about sellers, stock availability, and inventory levels.